

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 – ELECTROSTATICS (3.4) – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: How do invisible electric charges build up, exert forces across empty space, and store energy, and how can we harness these abilities to protect communities and power technology in Kenya? In this Final Explanation you will use evidence from all nine lessons to explain the lightning strike at Ol Kalou Market. Write in your own words, in full sentences, in the space under each prompt. Use evidence from the experiments you did (balloons, combs, electroscopes, the foil capacitor) and name the physics ideas you use. Include labelled diagrams or equations where they help. Refer back to the Ol Kalou market stall, the wooden stall and the lightning conductor on the district offices in every section.

Part 1: How Charge Builds Up and Behaves on Different Materials

Before the strike, the traders at Ol Kalou saw their arm hairs stand on end and heard their transistor radio crackle. (a) Using what you know about atoms, explain how objects become electrically charged and what the two types of charge are. (b) State the law of charges and use it to explain why the traders' arm hairs stood on end. (c) Use the idea of conductors and insulators to explain why the iron-roofed stall was struck but the wooden stall next to it was untouched. Support your answer with evidence from your friction and conductivity tests.

Write your answer here:

Part 2: Charging Without Touching - Induction and Earthing

The iron roof never touched the storm cloud, yet it gained charge. (a) Describe what you observed when a charged PVC pipe was brought near, without touching, the foil electroscope, and what happened when you touched the electroscope ball with your finger and removed the finger before removing the pipe. (b) Using these observations, explain step by step how the iron roof became charged by induction, including the sign of the charge on its upper surface. (c) Explain what earthing is, and how the lightning conductor on the district offices and the drag chain on a fuel tanker on Mombasa Road use it. (d) Explain why the wooden stall could not be charged by induction.

Write your answer here:

Part 3: Forces Across Empty Space - Coulomb's Law

Charged objects push and pull on each other without touching. (a) Describe what you observed about the force between two charged balloons as their distance changed, and what your F against $1/r^2$ graph showed. (b) Write Coulomb's law and explain what happens to the force if the distance is doubled, and if one of the charges is doubled. (c) Use Coulomb's law to explain why the iron roof, and not the wooden stall, was the structure the cloud acted on most strongly, and why the lightning conductor could intercept the discharge. (d) How does this idea help explain the force acting on charges in the traders' hairs and why the force acts across empty space?

Write your answer here:

Part 4: Storing and Releasing Energy - Capacitors

(a) Describe how you built and tested your foil capacitor, and what you observed when you changed plate area and plate separation. How did the LED show that energy had been stored? (b) Write the definition of capacitance and the formula for the energy stored in a capacitor. Explain, using the Q-V graph, why the energy has a factor of one half and why doubling the voltage has the effect it does. (c) Explain how the storm cloud and the iron roof acted like a capacitor, and use this to explain why the discharge could melt iron sheet. (d) Why does the lightning conductor provide a safer way to release this energy?

Write your answer here:

Part 5: Putting It Together - Explaining OI Kalou and Using Electrostatics in Kenya

Write a complete explanation of the OI Kalou Market lightning strike, in order from the build-up of charge to the final discharge. Your explanation must link charge separation, induction, Coulomb force, conductors and insulators, energy storage and the lightning conductor. Then choose one Kenyan application of electrostatics from the group presentations (for example electrostatic precipitators at Athi River, photocopiers in Kisumu, spray-painting booths on Mombasa Road or lightning conductors in Nyandarua) and explain which of the same ideas it uses to protect communities or power technology.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Cites specific observations from several lessons (balloons, comb, conductivity tests, electroscope, foil capacitor and LED) and connects each to the OI Kalou event.	Cites some evidence from the lessons and links most of it to the OI Kalou event.	Gives little or general evidence, or does not link it to the OI Kalou event.
Scientific accuracy of concepts	Correctly explains electron transfer, the law of charges, conductors and insulators, induction, earthing, Coulomb's law and capacitor energy with correct signs, formulas and units.	Explains most concepts correctly with small errors in wording, signs or units.	Shows major misconceptions, such as charges being created, or confuses the formulas.
Explaining the phenomenon as a cause-and-effect chain	Gives a complete, correctly ordered explanation from charge build-up to discharge that answers all the puzzles: hairs, radio, iron roof versus wooden stall, and lightning conductor.	Gives an ordered explanation that answers most of the puzzles but misses a link.	Lists ideas without a clear order or leaves most puzzles unanswered.
Quantitative reasoning and use of equations	States $F = kQ_1Q_2/r^2$, $C = Q/V$ and $E = 1/2 CV^2$ correctly and explains effects of doubling quantities, including the Q-V graph area.	States the equations correctly and explains most proportional effects.	Equations are missing or incorrect, or proportional effects are wrong.

Application to Kenyan technology and communication	Clearly links one Kenyan application to the same physics ideas, with clear writing and accurate scientific vocabulary.	Names an application and links it to some of the ideas, with generally clear writing.	Names an application without linking it to the physics, or writing is unclear.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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